METRIC
MIL-PRF-85045/36
18 May 2022

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, THIRTY-SIX SUBUNITS, TWELVE-FIBER ROLLED FLEXIBLE RIBBON SUBUNITS OR TWELVE-FIBER LOOSE TUBE SUBUNITS, ENHANCED PERFORMANCE, CABLE CONFIGURATION TYPE 4 (RIBBON CABLE) OR TYPE 1 (BUFFERED FIBER), CABLE CLASS SM AND MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

CLASSIFICATION:

Fiber optic cable configuration type:	Type 4 (ribbon cable) if each of the 36 subunits contain 12 fibers in a rolled flexible ribbon.
	Type 1 (buffered fiber) if each of the 36 subunits contain 12 loose (unbounded) fibers within a loose tube.
Fiber optic cable class:	MM (graded-index, glass core and glass cladding, multimode).
	SM (dispersion-unshifted, glass core and glass cladding, single mode).

DESIGN AND CONSTRUCTION:

Fiber:	MM fibers shall be in accordance with MIL-PRF-49291/6.
	SM fibers shall be in accordance with MIL-PRF-49291/7.

Buffer diameter. 250±15 micrometers (µm)

Fiber color coding. Each optical fiber in the subunit shall be color coded for identification in accordance with TIA-598 (see 3.4.2.4.1 of MIL-PRF-85045). A visual inspection performed after the temperature life test shall show that the colorant on each optical fiber remains intact with no flaking or other degradation in addition to no change outside specified limits for appearance (color) on the three samples of rolled flexible ribbon or loose tube subunit.

Subunit (loose tube or rolled flexible ribbon):

Dimensions and configuration: See figure 1 for rolled flexible ribbon and figure 2 for loose tube.

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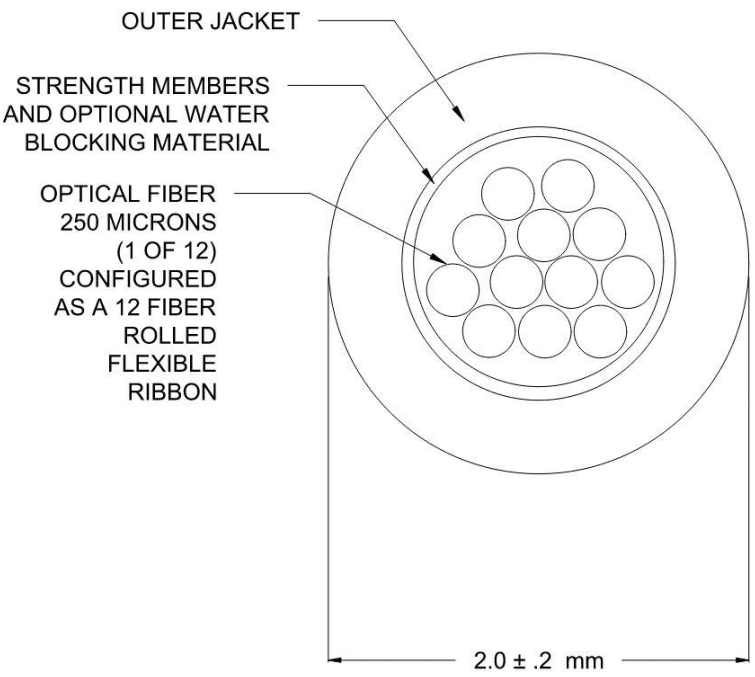


FIGURE 1. 12-fiber rolled flexible ribbon cable subunit.

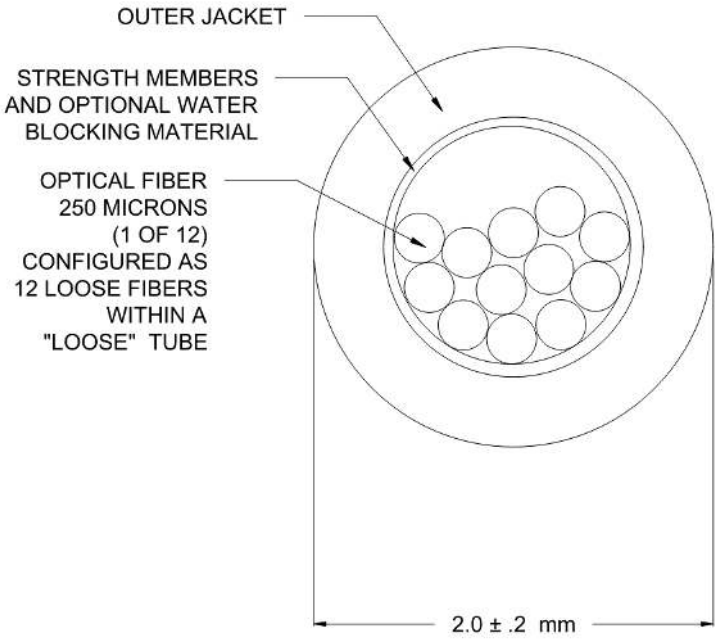


FIGURE 2. 12-fiber loose tube cable subunit.

Dimensions. See the requirement for the rolled flexible ribbon and loose tube outer diameter.

Fabrication. The ribbon shall have no crossovers. The ribbon or loose tube subunit shall have no defective fibers or splices.

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Mass per unit length. <5 kilograms per kilometer (kg per km).

Number of fibers. 12 (12 fibers in one rolled flexible ribbon or loose tube subunit).

Concentricity. ≥0.80.

Subunit cable jacket colors. Applicable (see 3.4.2.2.1 of MIL-PRF-85045).

Short-term minimum bend diameter. 36 times the subunit outer diameter.

Long-term minimum bend diameter. 16 times the subunit outer diameter.

Tensile loading. >270 newtons (N).

Dynamic bend tensile load. 90 N minimum.

Cable strain relief material requirement. The cable shall include a strength member (i.e., cable strain relief, such as one made from an aromatic polyimide, otherwise known as aramid yarn, or other solid form-of a liquid crystal polymer, such as an aromatic polyester). This cable strain relief material shall be capable of meeting the cable strain relief pullout force. Trade names for aramid yarns include Kevlar, Twaron, Nomex, New Star, and Teijinconex, or equal. Trade names for an aromatic polyester include Vectran, or equal.

Jacket red light transparency. Fiber breaks inside the subunit jacket shall be easily recognized when a pulsed, red light from a laser (monochromatic light) is sent through the optical fiber in the cable. The monochromatic light shall have a nominal wavelength of 635 nanometers (nm), a nominal pulse rate of 3 hertz, and a minimum optical power level of 1 milliwatt (mW). A fracture in the optical fiber shall be created by placing the middle of a 1-meter length of fiber optic cable in a free form loop until a kink occurs in the cable jacket and the change in optical transmittance of the cable is reduced by at least 20 decibels (dB).

Jacket material. The subunit jacket shall be composed of a low-halogen, low-smoke, low-toxicity polymer material.

Subunit outer diameter. The rolled flexible ribbon or loose tube shall be able to pass through a die with a 1.32±0.015-millimeter (mm) (0.052±0.0006-inch) diameter hole.

Rolled flexible ribbon subunit:

Configuration. The rolled flexible ribbon shall be configured as an adhesive matrix (bonded layer) applied over 12 adjacent optical fibers that are positioned laterally (in a linear array) with the adhesive matrix applied intermittently. At any cross section along the length, only two adjacent fibers shall be joined by the adhesive matrix (see “DEFINITIONS”).

Ribbon color coding. Not applicable for the ribbon adhesive matrix (see 3.4.2.4.2 of MIL-PRF-85045).

Tests for rolled flexible ribbon. Each test sample shall consist of 2 meters of finished cable or as specified in the test method. First, the specified testing shall be performed on the finished cable; then, the finished cable shall be deconstructed, and the group VII post-test examinations shall be performed on the rolled flexible ribbon lengths. Three samples shall be used for each test or as specified in the test method. Small end caps may be placed on the ends of the samples to eliminate end effects. A plastic cap or epoxy may be used. Test sample length and configuration for the ribbon twist and ribbon separation shall be as specified in those tests.

Ribbon strippability:

Method. One end of each rolled flexible ribbon shall be stripped to a minimum length of 30 mm with commercially available stripping tools to remove the binder (matrix) material and to remove the fiber coating (250-micron buffer). The test shall be performed both on unaged samples (from the ribbon dimension test) and on specimens of the rolled flexible ribbon from the finished cable that underwent temperature life testing followed by an inspection for ribbon delamination.

Requirement. No amount of fiber breakage in the rolled flexible ribbon shall occur and any residue of coating that remains shall be readily removed with isopropyl alcohol wipes.

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Ribbon twist (or twist robustness):

Method. Testing shall be performed on three samples of a 1-meter length of rolled flexible ribbon deconstructed from finished cable in accordance with TIA-455-141 for 20 cycles at a rate of 15±5 cycles per minute and at a rotation of 180±10 degrees in each direction for one cycle. The applied load shall have a weight of 500±25 grams. The cable length shall be sufficient to allow for a gauge length of 0.3±0.01 meter. A post-test visual inspection shall be performed.

Requirement. No amount of fibers shall become free and completely separate from the ribbon over the gauge length.

Ribbon delamination:

Method. Applicable except that three separate 2-meter samples of finished cable shall be included in the temperature life test versus performing a test under different conditions. After completion of the temperature test, the test samples of the rolled flexible ribbon (once the outer cable jacket and aramid yarn are removed) shall be inspected for ribbon delamination.

Requirement. A visual inspection performed after the temperature life test shall show no delamination occurred on the three samples of rolled flexible ribbon.

Ribbon separation:

Method. The optical fibers in a minimum of 0.5-meter length of rolled flexible ribbon shall be separated either by hand using no tools or by hand using tools without the aid of applied heat.

Requirement. The minimum 0.5-meter length of optical fibers in a rolled flexible ribbon shall be separated with no damage, removal, discoloration, or chemical attack to the 250-micron buffer. Minor removal of buffer colorant is acceptable.

Finished cable:

Dimensions and configuration. See figure 3. Thirty-six subunits shall be helically layered over the central member. The minimum outer jacket thickness shall be not less than 1.9 mm (establishes compatibility with termination and penetration devices). All subunits shall be of the same type (type 4 – rolled flexible ribbon or type 1 – loose tube). All 36 subunits shall be filled.

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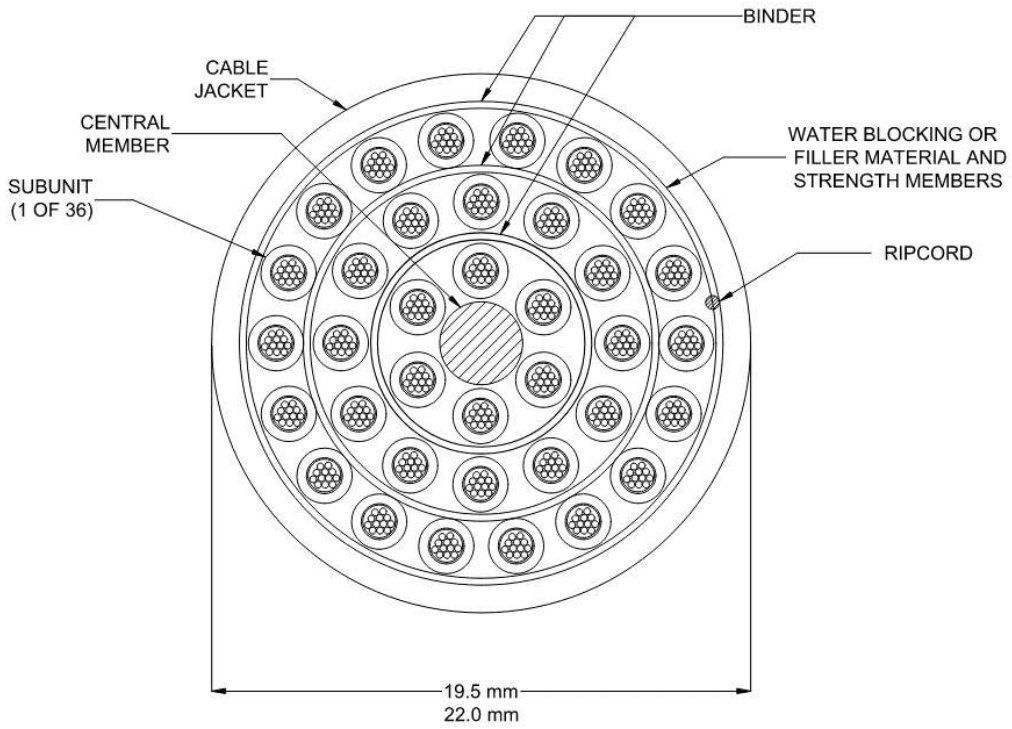


FIGURE 3. Multiple subunit, fiber optic cable with 36 subunits.

Number of fibers. 432 (12 per subunit).

Concentricity. >0.65.

Jacket material. The overall jacket shall be composed of a low-halogen, low-smoke, low-toxicity polymer material.

Mass per unit length. <640 kg per km.

Ripcord. A cord for opening the cable jacket along the length of the cable shall be provided as part of the finished cable. The ripcord shall run the entire length of the finished cable. Once a sufficient length of the cable outer jacket is removed to expose the interior cable elements, the ripcord shall be capable of opening the cable jacket along the cable length.

Short-term minimum bend diameter. Eight times the cable outer diameter. (The short-term minimum bend diameter shall be used in all environmental and mechanical tests that specify a cable minimum bend diameter.)

Long-term minimum bend diameter. 16 times the cable outer diameter.

Minimum continuous length. The minimum continuous length of all cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, a conformance inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

Cable jacket color. Shall be black or blue.

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PERFORMANCE REQUIREMENTS:

The individual performance requirements shall be as specified in MIL-PRF-85045 and this specification sheet.

Optical properties:

Maximum attenuation rate: 5.6 dB per km at 850±20 nm, 2.0 dB per km at 1300±20 nm for MM fiber.

1.5 dB per km at 1310±20 nm and 1550±20 nm for SM fiber.

For cables with radiation cross-linked jackets, the change in attenuation rate measurement may be made up to 30 days after cross-linking of the cable jacket.

Point discontinuity (attenuation uniformity):

Method. The attenuation uniformity of each individual fiber shall be measured in accordance with TIA-455-78. Unless otherwise specified, the attenuation uniformity shall be measured at a wavelength of 850 nm and 1300 nm for multimode and at 1310 nm and 1550 nm for single mode.

Requirement. There shall be no discontinuities in attenuation along the length greater than 0.1 dB for single mode and no discontinuities in attenuation along the length greater than 0.2 dB for multimode.

Bandwidth (multimode cables only). Applicable for cables with multimode fiber only. Fiber with a minimum bandwidth of 500 megahertz (MHz)-km at 1300 nm shall be used (multimode cables only). Bandwidth is not specified at 850 nm. Bandwidth shall be determined for the optical fiber and not the cable fabricated to this specification sheet (see “Intended use”).

Change in optical transmittance. Measurements shall be made at 1300±20 nm for multimode and 1310±20 nm and 1550±20 nm for single mode.

Measured fiber positions within the cable. The minimum number of active fibers required to be measured for change in optical transmittance during mechanical and environmental testing shall be as follows:

a. A minimum of 16 fibers shall be monitored for change in optical transmittance.

b. A minimum of four fibers shall be monitored in the inner ring of the subunits; each active fiber shall be from a different subunit.

c. A minimum of four fibers shall be monitored in the middle ring of the subunits; each active fiber shall be from a different subunit.

d. A minimum of eight fibers shall be monitored in the outer ring of the subunits; each active fiber shall be from a different subunit.

e. The fibers measured in the subunits for each ring shall be randomly selected, but different fibers shall be measured in each test sample.

For shock testing, only four fibers shall be required to be measured (one fiber for one subunit in the inner ring of subunits, one fiber for one subunit in the middle ring of subunits, and two fibers in two subunits in the outer ring of subunits), and the measurement shall be at 1310±20 nm for single mode.

Crosstalk. Crosstalk is applicable. For the optical measurement, a minimum of the two fibers adjacent to the active fiber shall be measured (e.g., if the active fiber is number 11, measure crosstalk on numbers 10 and 12). Repeat the process for a total of three active fibers per sample per unit.

Mechanical properties:

Tensile loading and elongation. Applicable; tensile loading >3,300 N.

Low temperature flexibility. The exposure temperature shall be -40 °C.

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Cyclic flexing. 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. A change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Cable twist-bending. 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. A change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Cable element removability. Cable element removability is applicable with the following additions:

a. Type 4 configuration only. The end of the rolled flexible ribbon shall be easily formed by hand into a flat ribbon that fits into an interfacing rectangular trough (such as tool [fusion splice holder], housing [splice tray], and component [MT terminus/connector]).

b. Type 4 configuration only. The 12 colored fibers shall be visually inspected to verify that they are retained in the specified order of color once flattened and the adhesive matrix is removed from the end of the 12 fibers.

c. Type 1 configuration only. The 12 colored fibers shall be visually inspected to verify that they can be placed and retained in the specified order of color and fitted into an interfacing rectangular trough (such as tool [fusion splice holder], housing [splice tray], and component [MT terminus/connector]).

d. The rip cord shall be capable of opening a 25-cm (10-inch) minimum length of cable jacket.

e. Holding the cable jacket (non-jacket opening end) is permissible during ripcord use to ensure that the ripcord does not slip out.

Impact. Applicable, except a third specimen of length 0.5 meter shall also be tested. The 0.5-meter specimen shall be tested at the low temperature only and the impact location shall be on the cable jacket over the rip cord location. No optical measurements are required for the 0.5-meter specimen.

Hosing. Low pressure is applicable, except the cable leakage shall be not greater than 135 milliliters.

Hydrostatic. 2.1 megapascals (MPa).

Cable scraping resistance. 750 cycles.

Cable-to-cable abrasion. 500 cycles.

Cable strain relief pullout force. The pullout force of the cable strain relief (i.e., the cable member for strain relief to a connection device comprised of a material such as aramid yarn) shall be performed on both the subunit (cable strain relief located within the subunit) and for the finished cable (cable strain relief located within the cable outer jacket).

a. Subunit. The pullout force of the cable strain relief shall be tested in accordance with TIA-455-6 by applying a 67-N (15-lb) load for a duration of 1 minute for a straight pull and a 33-N (7.5-lb) load for a duration of 1 minute at a 90-degree pull. The specimen shall consist of a commercial variant of the multi-fiber push on (MPO) connector with a 12-fiber MT ferrule and with a crimp barrel for a 2-mm crimp sleeve. The connector shall be affixed to the 2-meter cable sample at one end and a suitable connector to mate with the optical instrumentation detector interface at the other cable sample end. A test cable shall interface the specimen through a commercial variant of the MT ferrule at one cable end and the optical source connection at the other cable end. The epoxy used to secure the cable strain relief to the terminus shall be in accordance with two-part liquid epoxy as specified in appendix A of requirement 4102 of MIL-STD-1678-4.

b. Finished cable. The pullout force of the cable strain relief shall be tested in accordance with TIA-455-6 by applying the cable-to-backshell pullout force as specified in MIL-PRF-28876 for a duration of 10 minutes for a straight pull. One end on each length of fiber optic cable shall be terminated onto the end of a MIL-PRF-28876 connector with straight backshell (or other suitable fiber optic connector if approved by the qualifying activity prior to test sample fabrication). Fabrication shall be performed as specified in requirement 4102 of MIL-STD-1678-4.

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An axial tensile load shall be applied on the detector end cable (specimen) through a mandrel or other approved means. Load shall not be applied to the coupling mechanism (such as a MIL-PRF-28876 multiple termini connector). The change in optical transmittance shall be measured during and after the test and meet the optical requirement for this measurement. The cable strain relief shall be visually inspected after the test. No evidence of strain relief damage, breakage, or disengagement (pullout) shall occur.

Environmental properties:

Temperature range:

Operating: -28 to +65 °C

Nonoperating: -40 to +70 °C

Storage: -40 to +70 °C

Temperature cycling. A change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made over a period of 1 hour at the end of each temperature plateau.

Temperature humidity cycling. A change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Temperature life (life aging). Applicable, except that the jacket material shall be tested at 175 °C for 4 hours. A visual inspection performed after the temperature life test shall include inspection for the requirements specified under ribbon delamination and fiber color coding.

Fluid immersion. Exposure to automobile gasoline and tap water are not required and the following test temperatures shall be used for the fluids indicated: fuel oil (98 to 100 °C), turbine fuel (48 to 50 °C), and lubricating oil (98 to 100 °C).

Chemical properties:

Halogen content. <0.2 percent.

Cross-link verification (cables with cross-linked jackets only). The test shall be conducted in accordance with ICEA T-28-562 and run at 200 °C. The test shall be in the qualification test sequence using a specimen cut from the sample used in the thermal shock test and in the group IV testing of the group C conformance test sequence. The hot creep shall not exceed 100 percent, and the hot creep set shall not exceed 10 percent.

Identification markings. Identification markings are applicable and shall include the markings “432F” (for 432 fibers) and either “SM” (for single mode) or MM (for multimode), as appropriate.

Part or identifying number (PIN) (see table I):

M85045/36-01PM (multimode, 36 filled subunits with a 12-fiber rolled flexible ribbon in each subunit)

M85045/36-02PM (single mode, 36 filled subunits with a 12-fiber rolled flexible ribbon in each subunit)

M85045/36-03PM (multimode, 36 filled subunits with 12 fibers within the loose tube for each subunit)

M85045/36-04PM (single mode, 36 filled subunits with 12 fibers within the loose tube for each subunit)

“M” designates a 36-subunit cable.

“P” designates 7.7 MPa hydrostatic pressure proof cable.

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TABLE 1. Part or identifying number (PIN) designators.

PIN	Basic specification	Specification sheet	Designator: fiber type 1/, 2/	Designator: configuration type 3/	Designator: hydrostatic pressure 4/	Designator: number of subunits 5/
M85045/36-01PM	MIL-PRF-85045	/36	01	4	P	M
M85045/36-02PM	MIL-PRF-85045	/36	02	4	P	M
M85045/36-03PM	MIL-PRF-85045	/36	03	1	P	M
M85045/36-04PM	MIL-PRF-85045	/36	04	1	P	M

NOTES:

1/ A 36 subunit multiple subunit cable is a cable with 36 filled subunits. A filled subunit contains either a 12-fiber rolled flexible ribbon or 12 loose tube 250-micron buffered optical fibers.

2/ Fiber type designator:

01 = 36 filled ribbon fiber subunits with multimode fiber

02 = 36 filled ribbon fiber subunits with single mode fiber

03 = 36 filled loose tube subunits with multimode fiber

04 = 36 filled loose tube subunits with single mode fiber

3/ Configuration type designator:

Type 4 (ribbon cable) = each of the 36 subunits contain 12 fibers in a rolled flexible ribbon

Type 1 (buffered fiber) = each of the 36 subunits contain 12 fibers in a loose tube

4/ Hydrostatic pressure designator: “P” designates 7.7-MPa hydrostatic pressure-proof cable. There is no other option for this designator in this specification sheet.

5/ Number of subunit designator: “M” designates a 36-subunit cable. There is no other option (alternatives with some subunits containing other fiber sizes or no optical fibers) for this designator in this specification sheet.

Qualification and conformance inspection. See table II. This table specifies applicable requirements, testing, qualification inspections, and conformance inspections per this specification sheet. The requirement paragraph and test paragraph columns within this table reference MIL-PRF-85045. The cable length column of this table applies only to the qualification inspection. A reduced sample set is allowed for conformance inspection; see 4.6 of MIL-PRF-85045 for conformance inspection sample size.

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TABLE II. Qualification and conformance inspection.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length [1/, 2/]	Conformance inspection
I	Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	Three samples, 0.5 km each [3/]	A
	Attenuation rate	3.5.1	4.7.4.1	Three samples, 0.5 km each [4/]	A
	Point discontinuity	[5/]	[5/]	Three samples, 0.5 km each [4/]	--
II	Crosstalk	3.5.3	4.7.4.3	Three samples, 0.5 km each [4/]	--
	Electromagnetic effects	3.7.16	4.7.6.16	One specimen, 6 m [4/]	--
III	Temperature cycling	3.7.1	[5/]	Two samples, 0.5 km each [4/] (one on reel, one off)	C
	Temperature humidity cycling	3.7.3	4.7.6.3	Two samples, 0.5 km each [6/]	C
	Storage temperature	3.7.4	4.7.6.4	Two samples, 0.5 km each [6/]	--
	Low temperature flexibility (cold bend)	3.6.3	4.7.5.3	Two specimens, 8 m each [7/]	--
	Cyclic flexing	3.6.4	4.7.5.4	Six specimens, 5 m each [8/] (two specimens at each temperature)	--
	Crush	3.6.5	4.7.5.5	Two specimens, 5 m each [7/]	--
	Cable twist-bend	3.6.6	4.7.5.6	Six specimens, 5 m each [8/] (three specimens at each temperature)	C
	Impact	3.6.8	4.7.5.8	Three specimens, two at 5 m each and one at 0.5 m each [6/]	C
	Temperature life (life aging)	[5/]	[5/]	Two specimens, 300 m each [7/]	C
	Tensile loading and elongation	3.6.1	4.7.5.1	Two specimens, 150 m each [9/]	C
	Operating tensile loading	3.6.1.1	4.7.5.1.1	Two specimens, 150 m each [10/]	C
	Fungus resistance	3.8.4	4.8.4	Two specimens, 0.5 m each [7/]	--
	Cable element removability	3.6.18	4.7.5.18	Two specimens, 0.5 m each [7/]	C
IV	Thermal shock	3.7.2	4.7.6.2	One specimen, 0.49 km [4/] (on reel)	--
	Weathering	3.7.8	4.7.6.8	One specimen, 2 m [11/] and three material samples [12/]	--
	Fluid immersion	3.7.9	4.7.6.9	One specimen, 2 m [11/] and three material samples [12/] for each specified fluid	C
	Paint susceptibility	3.7.15	4.7.6.15	Two specimens, 2 m each [13/]	--

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TABLE II. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length [1/, 2/]	Conformance inspection
IV	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	One specimen, 30 m [11/]	--
	Shock	3.7.13	4.7.6.13	One specimen, 30 m [11/]	--
	Dynamic bend	3.6.2	4.7.5.2	One specimen, 150 m [14/]	B
	Hosing: low pressure	3.6.12.1	4.7.5.12.1	One specimen, 1.5 m [11/]	C
	Hosing: hydrostatic	3.6.12.2	4.7.5.12.2	One specimen, 2 m [11/]	C
	Radial compression (for application B only)	3.6.7	4.7.5.7	One specimen, 10 m [11/]	--
	Corner bend	3.6.9	4.7.5.9	Two specimens, 5 m each [13/]	--
	OFCC kink resistance	[5/]	[5/]	Two specimens, 1 m each [13/]	--
	Cross-link verification	[5/]	[5/]	Two specimens, 1 m each [13/]	C
V	Dripping	3.6.13	4.7.5.13	One specimen, 30 cm [11/]	--
	Cable jacket tear strength	3.6.14	4.7.5.14	Five flat extruded jacket material strips [15/]	C
	Cable jacket material tensile strength and elongation	3.6.15	4.7.5.15	Five specimens [16/]	--
	Cable abrasion resistance	3.6.16	4.7.5.16	Four specimens, 2 m each [17/]	C
	Cable shrinkage	3.6.17	4.7.5.17	Three specimens, 0.5 m each [13/]	C
	Durability of identification	3.6.19	4.7.5.19	Three specimens, 2 m each [13/]	--
	Flame extinguishing and smoke generation	3.7.12.2	4.7.6.12.2	One specimen, 50 m [11/]	C
	Water absorption	3.7.14	4.7.6.14	Two specimens, extruded jacket material strips [16/]	--
VI	Acid gas generation	3.8.1	4.8.1	One specimen, 1 m [18/]	C
	Halogen content	3.8.2	4.8.2	One specimen, 1 m [18/]	--
	Toxicity index	3.8.3	4.8.3	One specimen, 1 m [18/]	C
	Smoke index	3.8.5	4.8.6	One specimen, 1 m [18/]	C
VII	Ribbon strippability	[5/]	[5/]	Three specimens, 2 m each [18/]	--
	Ribbon twist	[5/]	[5/]	Three specimens, 1 m each [18/]	--
	Ribbon delamination	[5/]	[5/]	Three specimens, 2 m each [18/]	--
	Ribbon separation	[5/]	[5/]	One specimen, 0.5 m [18/]	--

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TABLE II. Qualification and conformance inspection – Continued.

NOTES:

1/ Tolerance on a 0.5-km length is ±5 percent, provided that results are normalized to 1 km.
2/ Tolerance on shorter lengths is ±5 percent.
3/ The visual and mechanical inspection shall only be conducted on a 2-meter section of each sample.
4/ The same samples as used in the visual and mechanical inspection shall be used.
5/ As stated under the applicable tests in this specification sheet.
6/ The same samples as used in the temperature cycling test shall be used.
7/ A specimen cut from each sample used in the temperature cycling test shall be used.
8/ Three specimens cut from each sample used in the temperature cycling test shall be used.
9/ A specimen cut from each specimen used in the temperature life test shall be used.
10/ The same specimen as used in the tensile loading and elongation test shall be used.
11/ A specimen cut from the specimen used in the thermal shock test shall be used.
12/ Three flat extruded jacket material strips as specified in 4.7.5.15 of MIL-PRF-85045 shall be used in this test.
13/ Specimens cut from the specimen used in the thermal shock test shall be used.
14/ A specimen cut from the specimen used in the thermal shock test or one of the specimens used in the tensile loading and elongation test shall be used.
15/ Flat extruded jacket material strips (i.e., strips of flat extruded material with the same composition as the cable jacket and of sufficient dimensions in which dumbbells can be cut) prepared as specified in 4.7.5.14 of MIL-PRF-85045 and obtained from flat extruded material that underwent the thermal shock test shall be used.
16/ Flat extruded jacket material as specified in 4.7.5.15 of MIL-PRF-85045 shall be used in this test.
17/ Specimen cut from the specimen used in the thermal shock test shall be used. Two specimens shall be used for scraping abrasion, and two specimens shall be used in the cable-to-cable abrasion testing.
18/ A specimen cut from one of the samples used in the temperature cycling test or specimen used in the thermal shock test shall be used.

Qualification sample configuration. For qualification only, each test sample may consist of mixed fibers for the same cable subunit configuration type (type 4 or type 1) in lieu of separate test samples for multimode fiber and for single mode fiber or mixed cable subunit construction (type 4 and type1) with the same fiber type (all multimode or all single mode). Each test sample with mixed fiber shall be restricted to the configuration of 18 subunits with multimode fiber and 18 subunits with single mode fiber. Allowance permission for the use of samples with mixed fiber is based on two constraints. The first constraint is that if one fiber type fails, then the cable fails (including the other fiber type). This constraint offsets added risk for reduced sample set for a lower qualification cost. The second constraint is if qualification testing is performed on MIL-PRF-85045/35 (8 subunit cables) first, then qualification testing may be performed on MIL-PRF-85045/36 with mixed fiber (18 subunits with multimode optical fiber, 18 subunits with single mode optical fiber). If only qualifying to MIL-PRF-85045/36 (36 subunit cable), separate qualifications shall be performed on MM and on SM cables. Allowance is only for cable configuration types (type 4 or type 1) that are the same.

Conformance sample configuration. The same sample configuration and constraint as for qualification applies.

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Qualification and conformance by similarity:

Qualification and conformance by similarity for single mode if qualified for MIL-PRF-85045/35. Manufacturers who produce products for both MIL-PRF-85045/35 and products for this specification sheet, are qualified under M85045/35-02P (or M85045/35-04P), and pass all applicable tests and inspections specified in table III are qualified under this specification sheet for M85045/36-02PM (or M85045/36-04PM). This qualification by similarity is applicable if the same materials are used in the previously qualified M85045/35-02P (or M85045/35-04P) cable and the cable under test and only for the same cable configuration types (type 4 or type 1).

TABLE III. Qualification and conformance by similarity to M85045/35.

Group	Inspection	Qualification inspection 1/	Conformance inspection 2/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
II	Crosstalk	X	--
	Electromagnetic effects	X	--
III	Temperature cycling	X	C
	Temperature humidity cycling	X	C
	Storage temperature	X	--
	Cyclic flexing	X	--
	Crush	X	--
	Cable twist-bending	X	C
	Impact (low temperature only)	X	C
	Temperature life (life aging)	X	C
	Tensile loading and elongation	X	C
	Operating tensile loading	X	C
	Cable element removability	X	C
IV	Thermal shock	X	--
	Shock	X	--
	Dynamic bend	X	B
	Hosing: low pressure	X	C
	Radial compression	X	---
	Corner bend	X	--
	OFCC kink	X	--
V	Cable abrasion resistance	X	C
	Cable shrinkage	X	C
	Flame extinguishing and smoke generation	X	C
	Water absorption	X	--
VI	Acid gas generation	X	C
	Toxicity index	X	C
	Smoke index	X	C

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TABLE III. Qualification and conformance by similarity to M85045/35 – Continued.

NOTES:

1/ Where manufacturers are qualified for M5045/35-02P and are pursuing qualification for M85045/36-02PM or the manufacturers are qualified for M85045/35-04P and are pursuing qualification for M85045/36-04PM.

2/ Where manufacturers are qualified for M85045/35-02P and M85045/36-02PM (or M85045/35-04P and M85045/36-04PM, respectively) and are performing conformance testing for M85045/36-02PM (or M85045/36-04PM). Allowance is only for cable configuration types (type 4 or type 1) that are the same.

Qualification and conformance by similarity for a change in fiber type. Manufacturers who are qualified under this specification sheet for single mode fiber cable, M85045/36-02PM (or M85045/36-04PM), and whose multimode fiber cable passes all tests and inspections specified in table IV shall be qualified under this specification sheet for multimode fiber cable, M85045/36-01PM (or M85045/36-03PM). This qualification by similarity shall be applicable if the only difference between the previously qualified cable M85045/36-02PM (or M85045/36-04PM) and the cable under test is that the optical fiber has been changed from a single mode fiber to a multimode fiber for the same cable configuration type (type 4 or type 1). Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order shall be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

TABLE IV. Qualification and conformance by similarity (fiber).

Group	Inspection	Qualification inspection M85045/36-01PM (or M85045/36-03PM) 1/	Conformance inspection M85045/36-01PM (or M85045/36-03PM) 2/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Temperature cycling	X	C
	Temperature humidity cycling	X	C
	Storage temperature	X	--
	Cyclic flexing	X	--
	Crush	X	--
	Cable twist-bending	X	C
	Impact (low temperature only)	X	C
	Tensile loading and elongation	X	C
	Operating tensile loading	X	C
IV	Thermal shock	X	--
	Dynamic bend	X	B

NOTES:

1/ Where manufacturers are qualified for M85045/36-02PM and are pursuing qualification for M85045/36-01PM or the manufacturers are qualified for M85045/36-04PM and are pursuing qualification for M85045/36-03PM.

2/ Where manufacturers are qualified for M85045/36-01PM (or M85045/36-03PM) and M85045/36-02PM (or M85045/36-04PM) and are performing conformance testing for M85045/36-01PM (or M85045/36-03PM). Allowance is only for cable configuration types (type 4 or type 1) that are the same.

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Different cable configuration type by similarity (type 4 and type 1). Manufacturers who are qualified under this specification sheet for single mode fiber cable (M85045/36-02P) with a 12-fiber rolled flexible ribbon in each subunit and whose single mode fiber cable (M85045/36-04P) with 12 loose fibers within the loose tube for each subunit passes the tests and inspections specified in table V are qualified under this specification sheet for single mode fiber cable (M85045/36-04P). This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the cable configuration type had been changed from cable configuration type 4 to cable configuration type 1. Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. The test order shall be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

TABLE V. Different cable configuration type by similarity (fiber).

Group	Inspection	Qualification inspection M85045/34-04P 1/	Conformance inspection M85045/34-04P 2/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Temperature cycling	X	C
	Temperature humidity cycling	X	--
	Low temperature flexibility (cold bend)	X	B
	Cyclic flexing	X	--
	Crush	X	--
	Cable twist-bending	X	C
	Impact (low temperature only)	X	C
	Tensile loading and elongation	X	C
	Operating tensile loading	X	C
IV	Thermal shock	X	--
	Dynamic bend	X	B
	Hosing: low pressure	X	C
	Hosing: hydrostatic	X	--
	Kink resistance	X	--
V	Cable shrinkage	X	--
	Flame extinguishing and smoke generation	X	--
	Cable strain relief pullout force	X	--
VI	Halogen content	X	--
	Smoke index	X	C

NOTES:

1/ Where manufacturers are qualified for M85045/36-02P and are pursuing qualification for M85045/36-04P.

2/ Where manufacturers are qualified for M85045/36-04P and are performing conformance testing for M85045/36-04P.

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Intended use. This cable is intended for use in cable runs between fiber optic switches (backbone cables) and in cable runs to end-user equipment (cable drops). This cable is intended for use outside of protected enclosures such as for installation in the overheads or cableways outside protective enclosures. This multiple subunit fiber optic ribbon cable is used when the affixed connector insert contains multiple MT ferrules and used for high density cabling applications where cable ends are joined by fusion splicing.

Bandwidth measurements (multimode cables only). For this cable construction, the bandwidth is measured as part of the requirements of the optical fiber to be cabled. The bandwidth value does not change once the optical fiber is cabled. Since the bandwidth is provided as one of the cable-measured parameters, the acceptable bandwidth for the optical fiber used in this cable (and by extension, the cable) is provided for informational purposes. For a cable with MM 62.5/125-micron fiber: ≥200 MHz-km at 850±20 nm, ≥500 MHz-km at 1300±20 nm for cable with MM fiber.

DEFINITIONS:

Rolled. The term rolled is used to emphasize the requirement for the ribbon to be rolled longitudinally within the cable to increase the fiber packing density (allow for a specified number of fibers for the given cable diameter).

Flexible. The term flexible is used to emphasize the requirement that the application of the adhesive matrix is applied intermittently to allow the rolled flexible ribbon to take a planar shape for connector termination and splicing and take a circular shape when rolled longitudinally within the cable.

Referenced documents. In addition to MIL-PRF-85045, this document references the following:

MIL-PRF-28876	MIL-STD-1678-4	TIA-598
MIL-PRF-49291/6	TIA-455-6	ICEA-T-28-562
MIL-PRF-49291/7	TIA-455-78	
MIL-PRF-85045/35	TIA-455-141	

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
NASA – NA

Preparing activity:
Navy – SH
(Project 6015-2021-006)

Review activities:
Army – AR, AV, MI
Navy – AS, EC, OS, YD
Air Force – 02, 13, 19, 70, 93
DLA – CC
MISC – MDA

Civil agency:
GSA – FAS

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